

Basic Array Questions

1. Print all elements of an array

Question:

Given an array `arr[]`, print all elements.

Example:

```
Input: [1,2,3,4]
Output: 1 2 3 4
```

Concept:

- Traversal
- for loop

2. Find the sum of all elements

Question:

Return the sum of all elements in an array.

Example:

```
Input: [2,4,6]
Output: 12
```

Concept:

- Accumulator pattern

3. Find the largest element

Question:

Find the maximum number in an array.

Example:

```
Input: [3,9,2,7]
Output: 9
```

Concept:

- Comparison
 - Updating variables
-

4. Find the smallest element

Question:

Find the minimum element in an array.

Example:

```
Input: [7,2,9,1]
Output: 1
```

Concept:

- Tracking minimum
-

5. Search an element (Linear Search)

Question:

Check whether a given number exists.

Example:

```
Input: [3,6,9], target=6
Output: Found
```

Concept:

- Searching
 - Condition checking
-

6. Count even and odd numbers

Question:

Count how many even and odd numbers exist.

Example:

```
Input: [1,2,3,4]
Output:
Even = 2
Odd = 2
```

Concept:

- Modulus %
-

7. Print alternate elements

Question:

Print elements at index 0, 2, 4...

Example:

```
Input: [1,2,3,4,5]
Output: [1,3,5]
```

Concept:

- Step increment (i += 2)
-

8. Reverse an array

Question:

Print array in reverse order.

Example:

```
Input: [1,2,3]
Output: [3,2,1]
```

Concept:

- Reverse traversal
-

9. Count frequency of a number

Question:

Count how many times a given number appears.

Example:

```
Input: [1,2,1,3,1], target=1  
Output: 3
```

Concept:

- Counting pattern
-

10. Find second largest element

Question:

Find second biggest number.

Example:

```
Input: [10,4,20,15]  
Output: 15
```

Concept:

- Multiple variable tracking
 - Logic building
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Rule while solving:

1. Dry run on paper
2. Write loop
3. Write condition
4. Test with small input